

## FUKUE, Japan

WMO: 478430

Lat: 32.6947N

Lon: 128.8269E

Elev: 88

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 33.7       | 35.7 | 20.5                          | 15.4 | 39.3 | 22.5 | 17.0 | 40.5 | 21.2                  | 43.3 | 19.5 | 44.3 | 7.1                   | 210  | 0.394 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 10.6                   | 89.3            | 78.9 | 87.8 | 78.6 | 86.4 | 78.2 | 80.8                | 85.9 | 80.0 | 85.0 | 79.3 | 84.2 | 8.8                  | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.5                            | 153.6 | 83.6 | 78.7 | 149.6 | 83.2 | 78.0 | 145.7 | 82.8 | 44.6          | 86.3 | 43.8 | 85.1 | 43.0 | 84.2 | 83.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 20.1              | 17.6 | 15.7 | DB                         | 30.0 | 92.3               | 1.9 | 1.9                                                | 28.6 | 93.6       | 27.5 | 94.7       | 26.5 | 95.8       | 25.1 | 97.2 |
|                   |      |      | WB                         | 27.7 | 81.5               | 1.6 | 1.3                                                | 26.5 | 82.5       | 25.5 | 83.2       | 24.6 | 83.9       | 23.4 | 84.8 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 62.7   | 45.2  | 46.6  | 51.2  | 58.6  | 65.9  | 71.9  | 79.7  | 81.6  | 75.8  | 67.6  | 58.1  | 49.3  | (6)  |
|                                                                        | DBStd       | 13.19  | 4.90  | 5.45  | 5.17  | 4.73  | 3.44  | 3.37  | 3.51  | 2.75  | 3.80  | 4.35  | 5.49  | 5.25  | (7)  |
|                                                                        | HDD50       | 413    | 163   | 122   | 46    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 77    | (8)  |
|                                                                        | HDD65       | 2506   | 613   | 514   | 428   | 197   | 30    | 0     | 0     | 0     | 0     | 23    | 215   | 486   | (9)  |
|                                                                        | CDD50       | 5060   | 15    | 28    | 83    | 260   | 494   | 657   | 922   | 980   | 774   | 545   | 246   | 56    | (10) |
|                                                                        | CDD65       | 1678   | 0     | 0     | 0     | 5     | 59    | 207   | 457   | 515   | 324   | 103   | 8     | 0     | (11) |
|                                                                        | CDH74       | 13271  | 0     | 0     | 0     | 2     | 133   | 803   | 4320  | 5533  | 2171  | 303   | 6     | 0     | (12) |
|                                                                        | CDH80       | 3582   | 0     | 0     | 0     | 0     | 3     | 39    | 1177  | 1912  | 436   | 15    | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 7.0    | 7.8   | 7.9   | 8.1   | 7.2   | 6.3   | 6.0   | 7.0   | 6.4   | 6.4   | 6.8   | 6.8   | 7.3   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 89.6   | 3.8   | 4.4   | 6.5   | 8.6   | 8.5   | 12.9  | 11.1  | 9.1   | 10.1  | 4.8   | 5.5   | 4.2   | (15) |
|                                                                        | PrecMax     | 119.6  | 9.1   | 8.8   | 15.7  | 16.0  | 15.4  | 25.6  | 19.7  | 19.9  | 24.1  | 12.8  | 13.5  | 8.9   | (16) |
|                                                                        | PrecMin     | 61.6   | 0.5   | 0.7   | 2.3   | 3.8   | 3.6   | 2.1   | 1.2   | 2.4   | 2.7   | 0.1   | 0.2   | 0.6   | (17) |
|                                                                        | PrecStd     | 14.4   | 2.0   | 2.0   | 2.7   | 3.2   | 3.4   | 5.5   | 5.0   | 4.0   | 5.6   | 3.6   | 3.3   | 2.3   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 63.4  | 65.7  | 68.1  | 73.5  | 79.9  | 83.2  | 90.5  | 91.9  | 88.0  | 81.9  | 74.6  | 66.8  | (19) |
|                                                                        |             | MCWB   | 59.5  | 61.6  | 61.0  | 63.0  | 64.8  | 73.7  | 79.0  | 79.3  | 77.3  | 73.8  | 66.7  | 61.6  | (20) |
|                                                                        | 2%          | DB     | 59.7  | 62.8  | 65.4  | 71.0  | 77.2  | 81.0  | 88.5  | 90.1  | 86.1  | 79.2  | 72.0  | 63.7  | (21) |
|                                                                        |             | MCWB   | 55.0  | 58.7  | 59.2  | 62.4  | 64.6  | 73.0  | 78.9  | 79.0  | 77.0  | 71.3  | 65.0  | 57.5  | (22) |
|                                                                        | 5%          | DB     | 56.6  | 59.8  | 63.6  | 69.2  | 75.3  | 79.6  | 87.1  | 88.7  | 84.3  | 77.3  | 69.8  | 61.2  | (23) |
|                                                                        |             | MCWB   | 50.9  | 54.5  | 58.0  | 61.9  | 64.8  | 72.7  | 78.7  | 78.8  | 75.9  | 69.0  | 63.3  | 54.9  | (24) |
|                                                                        | 10%         | DB     | 53.9  | 56.5  | 61.4  | 67.4  | 73.5  | 78.3  | 85.7  | 87.4  | 82.6  | 75.4  | 67.6  | 58.9  | (25) |
|                                                                        |             | MCWB   | 48.0  | 50.7  | 55.6  | 60.6  | 64.7  | 72.3  | 78.3  | 78.5  | 74.8  | 67.2  | 61.2  | 52.6  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 60.6  | 62.6  | 63.7  | 68.7  | 72.9  | 78.1  | 81.8  | 81.9  | 80.0  | 76.9  | 69.8  | 64.0  | (27) |
|                                                                        |             | MCDB   | 62.6  | 64.7  | 65.5  | 70.3  | 74.3  | 80.3  | 86.9  | 87.6  | 84.3  | 79.8  | 73.1  | 66.0  | (28) |
|                                                                        | 2%          | WB     | 56.1  | 59.5  | 61.5  | 66.3  | 70.5  | 76.7  | 80.8  | 81.0  | 78.8  | 73.9  | 67.0  | 59.1  | (29) |
|                                                                        |             | MCDB   | 58.8  | 62.3  | 64.2  | 69.1  | 72.4  | 79.3  | 85.3  | 86.5  | 83.4  | 77.8  | 70.3  | 62.2  | (30) |
|                                                                        | 5%          | WB     | 52.1  | 55.5  | 59.1  | 64.2  | 68.7  | 75.6  | 80.0  | 80.2  | 77.6  | 71.3  | 64.8  | 56.1  | (31) |
|                                                                        |             | MCDB   | 55.5  | 58.7  | 62.7  | 67.2  | 71.8  | 78.3  | 84.4  | 85.7  | 82.4  | 75.3  | 68.6  | 59.9  | (32) |
|                                                                        | 10%         | WB     | 48.7  | 51.7  | 56.5  | 62.2  | 67.0  | 74.3  | 79.2  | 79.7  | 76.5  | 68.7  | 62.2  | 53.2  | (33) |
|                                                                        |             | MCDB   | 53.0  | 55.6  | 60.6  | 65.8  | 71.3  | 77.1  | 83.6  | 85.1  | 81.2  | 73.7  | 66.7  | 58.3  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.6  | 11.9  | 13.4  | 13.9  | 13.6  | 10.5  | 9.3   | 10.6  | 11.1  | 12.4  | 12.8  | 11.6  | (35) |
|                                                                        |             | MCDBR  | 16.2  | 16.2  | 16.4  | 16.2  | 17.0  | 12.4  | 11.4  | 12.3  | 12.4  | 14.0  | 14.5  | 15.5  | (36) |
|                                                                        |             | MCWBR  | 12.7  | 13.4  | 12.5  | 10.4  | 9.1   | 6.4   | 4.8   | 4.9   | 6.0   | 7.8   | 10.1  | 11.2  | (37) |
|                                                                        | 5% WB       | MCDBR  | 14.8  | 14.7  | 14.6  | 11.6  | 10.6  | 7.3   | 9.6   | 10.6  | 10.6  | 11.0  | 12.3  | 13.8  | (38) |
|                                                                        |             | MCWBR  | 13.8  | 14.2  | 13.5  | 9.9   | 8.4   | 5.1   | 4.3   | 4.9   | 5.8   | 7.7   | 10.4  | 12.2  | (39) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.455 | 0.489 | 0.544 | 0.559 | 0.564 | 0.577 | 0.555 | 0.524 | 0.501 | 0.468 | 0.442 | 0.442 | (40) |
|                                                                        | taud        |        | 2.007 | 1.953 | 1.830 | 1.849 | 1.897 | 1.960 | 2.059 | 2.162 | 2.163 | 2.139 | 2.131 | 2.046 | (41) |
|                                                                        | Ebn at Noon |        | 231   | 237   | 235   | 238   | 239   | 234   | 239   | 245   | 245   | 242   | 234   | 226   | (42) |
|                                                                        | Edh at Noon |        | 45    | 53    | 64    | 65    | 63    | 59    | 53    | 47    | 45    | 43    | 40    | 41    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 638   | 896   | 1270  | 1532  | 1631  | 1398  | 1706  | 1744  | 1384  | 1157  | 800   | 594   | (44) |
|                                                                        | RadStd      |        | 71    | 108   | 93    | 131   | 197   | 207   | 229   | 149   | 131   | 109   | 86    | 53    | (45) |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | +0.76 | +1.01 | N/A         | N/A   | N/A   | N/A   |
| Regional (14 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page